

Review

47%



7

Out of 15 points

1:29

Avg Time Per Question

Your Answers:

Vignette

An equity index is established in Year 1 for a country that has recently moved to a market economy. The index vendor constructed returns for the four years prior to Year 1 based on the initial group of companies constituting the index in Year 1. From Year 12 to Year 16, a series of military confrontations concerning a disputed border disrupted the economy and financial markets. The dispute is conclusively arbitrated at the end of Year 16. In total, 20 years of equity market return history is available. Other selected data are in the following tables.

Selected Data

Geometric mean return relative to 10-year government bond returns (over a 20-year period)	2% per year
Arithmetic mean return relative to 10-year government bond returns (over a 20-year period)	2.3% per year
Index forward dividend yield	1%
Forecasted public company earnings growth	5% per year
Forecasted market P/E growth	1% per year
Forecasted real GDP growth rate (by Year 19)	4%
Current vs. long-term inflation forecast	6% vs. 4% per year

Current yield curve (inversion)

Short maturities:
9%

10-year
maturities: 7%

1 Multiple Choice 1 / 1 point

The inclusion of index returns prior to Year 1 would be expected to:

☒ A. bias the historical ERP estimate upward.

☐ B. bias the historical ERP estimate downward.

☐ C. have no effect on the historical ERP estimate.

Feedback

A is correct. The backfilling of index returns using companies that have survived to the index construction date is expected to introduce a positive survivorship bias into returns.

2 Multiple Choice 0 / 1 point

The events of 2012 to 2016 would be expected to:

☒ A. bias the historical ERP estimate upward.

Correct Answer: **B. bias the historical ERP estimate downward.**

☐ B. bias the historical ERP estimate downward.

☐ C. have no effect on the historical ERP estimate.

Feedback

B is correct. The events of Year 12 through Year 16 depressed share returns but (1) are not a persistent feature of the stock market environment, (2) were not offset by other positive events within the historical record, and (3) have led to relatively low valuation levels, which are expected to rebound.

3 Multiple Choice 0 / 1 point

In the current interest rate environment, using a required return on equity estimate based on the short-term government bond rate and a historical ERP defined in terms of a short-term government bond rate would be expected to:

☐ A. bias long-term required return on equity estimates upward.



☒ B. bias long-term required return on equity estimates downward.

Correct Answer:

A. bias long-term required return on equity estimates upward.

☐ C. have no effect on long-term required return on equity estimates.

Feedback

A is correct. The required return reflects the magnitude of the historical ERP, which is generally higher when based on a short-term interest rate (as a result of the normal upward-sloping yield curve), and the current value of the rate being used to represent the risk-free rate. The short-term rate is currently higher than the long-term rate, which will also increase the required return estimate. The short-term interest rate, however, overstates the long-term expected inflation rate. Using the short-term interest rate, estimates of the long-term required return on equity will be biased upward.

4 Multiple Choice 1 / 1 point

An estimate of the ERP consistent with the Grinold–Kroner model is *closest* to:

☐ A. 2.7%.



☒ B. 3.0%.

☐ C. 4.3%.

Feedback

B is correct.

$i = 4\%$ per year (long-term forecast of inflation)

$g = 4\%$ per year (growth in real GDP)

$\Delta(P/E) = 1\%$ per year (growth in market P/E)

$dy = 1\%$ per year (dividend yield or the income portion)

Risk-free return = $r_f = 7\%$ per year (for 10-year maturities)

Using the Grinold–Kroner model, the ERP estimate is

$$\text{ERP} = \{1.0 + 1.0 + [4.0 + 4.0 + 0.0]\} - 7.0 = 3.0\%.$$

The premium of 3.0% compensates investors for average market risk, given expectations for inflation, real earnings growth, P/E growth, and anticipated income.

5 Multiple Choice 0 / 1 point

Common stock issues in the aforementioned market with average systematic risk are *most likely* to have required rates of return of:

- ☐ A. between 2% and 7%.
- ☒ B. between 7% and 9%.

Correct Answer: **C. 9% or greater.**

- ☐ C. 9% or greater.

Feedback

C is correct. Based on a long-term government bond yield of 7%, a beta of 1, and any of the risk premium estimates that can be calculated from the givens (e.g., a 2% historical risk premium estimate or 3.0% Grinold–Kroner ERP estimate), the required rate of return would be at least 9%. Based on using a short-term rate of 9%, C is the correct choice.

6 Multiple Choice 1 / 1 point

An analyst wants to use a forward-looking approach to estimate the equity risk premium of a market in which public equities represent a relatively large share of the economy. The analyst has collected inflation and earnings growth data and wants to avoid the estimate being sensitive to recent market returns. The analyst also believes it is important to account for different rates of growth in earnings, dividends, and prices. The analyst should use which of the following estimation methods?

- ☐ Survey-based estimates

Incorrect Answer Feedback: Incorrect because surveys used to estimate the equity risk premium tend to be sensitive to recent market returns. Because the analyst wants to avoid his estimate being sensitive to recent market returns, survey-based estimates is not the best method to use.

☒ Macroeconomic modeling

Correct Answer Feedback: Correct because equity risk premium estimates derived from macroeconomic models rely on a number of forecasted economic variables such as inflation and expected growth in real earnings per share. These models might be more reliable when public equities represent a relatively large share of the economy, as in many developed markets. Given the analyst's use of inflation and growth data, coupled with the fact that public equities represent a relatively large share of the economy, macroeconomic modeling is the best choice.

☐ The Gordon growth model

Incorrect Answer Feedback: Incorrect because the Gordon growth model does not allow for changes in growth, dividends, and prices. An underlying assumption of the Gordon growth model is that earnings, dividends, and prices will grow at the same rate, resulting in a constant P/E. If, however, the analyst believes this is not likely to be the case going forward, an adjustment would be needed that reflects anticipated P/E multiple expansion or contraction. Therefore, using the Gordon growth model is not useful because the analyst believes that it is important to allow for different rates for the growth earnings, dividends, and prices in the estimation.

7 Multiple Choice 0 / 1 point

Last year, a company acquired an asset using a finance lease. At that time, the company disclosed the interest rate for the lease liability and the rate of interest it would pay to borrow, using a collateralized loan over the same term. However, the company did not disclose the direct initial costs associated with the lease. The company has no other liabilities. Since then, interest rates have significantly increased. Which of the following is the *most* appropriate method to estimate the company's current cost of debt?

☒ Calculate the rate implicit in the lease

Incorrect Answer Feedback: Incorrect because the rate implicit in the lease (RIIL) is the discount rate that equates the sum of the present value of the lease payments and the present value of the residual value with the sum of the fair value of the leased asset and the lessor's direct costs (e.g., legal fees). However, the direct costs associated with the lease are not disclosed. Therefore, the RIIL cannot be calculated.

☒ Determine the company's incremental borrowing rate

Correct Answer Feedback: Correct because the present value of the residual value and the lessor's direct initial costs are often not known to the lessee (company) or analyst. If unknown [as in this case], the incremental borrowing rate (IBR), which is the rate of interest the company would pay to borrow using a collateralized loan over the same term, might be used. Therefore, provided the change in the interest environment, the IBR is an appropriate method to estimate the cost of debt of the company.

☐ Use the interest rate disclosed by the company at the time of asset acquisition

Incorrect Answer Feedback: Incorrect because since the time of asset acquisition, interest rates have significantly increased. Therefore, using the interest rate disclosed at time of asset acquisition is not an accurate estimator for the current cost of debt of the company.

8 Multiple Choice 0 / 1 point

Which of the following is a bottom-up factor affecting a company's cost of capital?

☐ ESG risk



Correct Answer Feedback: Correct because ESG risks are specific to an individual company and are therefore bottom-up factors. Even if industry peers face similar environmental risks, they will have varied approaches to dealing with them. No two companies are exactly alike with respect to their ESG risks, as reflected in company disclosures. A company with higher environmental, social, and governance (ESG) risk is likely to have a higher cost of capital.

☐ The risk-free rate



Incorrect Answer Feedback: Incorrect because the risk-free rate is a macroeconomic factor and therefore a top-down factor. Macroeconomic and country-specific economic factors, such as inflation rates, are reflected in benchmark interest rates and the overall level of credit spreads. Higher relative rates of inflation, represented in a higher risk-free rate, increase the cost of capital for companies.

☐ Capital availability

Incorrect Answer Feedback: Incorrect because capital availability represents a top-down, and not a bottom-up factor, that influences a company's cost of capital.

9 Multiple Choice 1 / 1 point

The equity risk premium (ERP) demanded by investors will *most likely*:

☐ decrease during recessions.

Incorrect Answer Feedback: Incorrect because a company's cost of capital is highly influenced by market conditions such as interest rates, inflation rates, and the macroeconomic environment. The ERP demanded by investors tends to increase during recessions and decrease during expansionary times, not vice versa.

☐ move inversely with relative rates of inflation.

Incorrect Answer Feedback: Incorrect because macroeconomic and country-specific economic factors, such as inflation rates. When interest rates are relatively low and credit spreads are tight, the costs of debt and equity capital are lower. Higher relative rates of inflation, represented in a higher risk-free rate, increase the cost of capital for companies. Thus, as relative rates of inflation increase, the equity risk premium demanded by investors would also most likely increase.

☒ be lower in developed economies than in developing economies.



Correct Answer Feedback: Correct because developed economies typically have more established, liquid capital markets with greater capital availability, more stable currencies, better property protection, and a greater strength in the rule of law than those of developing economies. Consequently, the perceived risk associated with investing in companies in more mature capital markets is lower than for companies in less mature economies. Lower perceived risk translates into lower credit spreads, ERPs, and costs of capital for companies in more mature, or developed, economies.

10 Multiple Choice 1 / 1 point

Which of the following models for estimating a company's required return on equity does **not** directly account for a size premium?

☐ Expanded CAPM

Incorrect Answer Feedback: Incorrect because the expanded CAPM model accounts directly for the size premium when estimating the required return on the company's equity. To estimate r_e for non-publicly traded or private companies, analysts can use an adaptation of the CAPM called the expanded CAPM, which adds a premium for small company size and other company-specific risks. The expanded CAPM requires estimation of a beta from a peer group of publicly traded companies.

☐ Three-factor Fama–French model

Incorrect Answer Feedback: Incorrect because the three-factor Fama–French model accounts directly for the size premium when estimating the required return on the company's equity. In the Fama–French three-factor model, in addition to the single market factor, equity returns can be explained by the size of the company—a size factor measured by market capitalization—and the relationship between the book value and equity value of a company's equity, termed the value factor. Using this three-factor model, a company's excess return on equity is calculated as:

$$r_e = r_f + \beta_1 \text{ERP} + \beta_2 \text{SMB} + \beta_3 \text{HML}$$

where SMB is the size premium, equal to the average difference in equity returns between companies with small and large capitalizations, and HML is the value premium, equal to the average difference in equity returns between companies with high and low book-to-market ratios.

☒ Bond yield plus risk premium model

Correct Answer Feedback: Correct because the bond yield plus risk premium (BYPRP) approach estimates a company's required return on equity as:

$$r_e = r_d + \text{RP}$$

where r_d is the company's cost of debt (typically proxied by the YTM on the company's long-term debt), and RP is a risk premium to compensate equity investors for additional risk relative to the risk of investing in the company's debt securities. This model does not directly account for the size premium as the factor.

11 Multiple Choice 1 / 1 point

An analyst gathers the following data about a company's publicly traded preferred equity:

Dividend rate	6.0%
Par value	\$1100
Current share price	\$990

The current cost of the preferred equity is *closest* to:

☐ 5.4%.

Incorrect Answer Feedback: Incorrect because this answer mixes current price and par value, and calculates annual dividend as current share price multiplied by dividend rate ($\$59.4 = 990 \times 0.06$), and then cost of equity as $r_e = 59.4 / 1100 + 0 = 5.4\%$.

☐ 6.0%.

Incorrect Answer Feedback: Incorrect because this answer is simply the dividend rate of the preferred equity and does not take into account the difference between the par value and the current share price. Since the stock is actively traded, the current market price should be used to determine the cost of preferred equity.

☒ 6.7%.

Correct Answer Feedback: Correct because based on dividend discount model, required return on equity (or cost of equity) can be calculated as:

$$r_e = D_1 / P_0 + g$$

where D_1 is future expected annual dividend, P_0 is the current share price, and g is expected growth rate in dividends. Based on the data above, $D_1 = \text{par value} \times \text{dividend rate} = \$1100 \times 0.06 = \$66$, and $g = 0$, as the dividend is expected to be flat. Thus, $r_e = 66 / 990 + 0 = 6.67\% \approx 6.7\%$.

12 Multiple Choice 0 / 1 point

An analyst wants to estimate the cost of equity of a private company using the build-up approach, and gathers the following data about relevant risk premia:

Equity risk premium (ERP)	6.0%
Size premium (SP)	4.5%
Specific-company risk premium (SCRCP)	5.0%

If the risk-free rate is 3.5% and peer beta is 1.5, the estimated cost of equity is *closest* to:

☐ 15.5%.

Incorrect Answer Feedback: Incorrect because this answer choice omits the risk-free rate:

$$r_e = 6.0\% + 4.5\% + 5.0\% = 15.5\%.$$

☒ 19.0%.

Correct Answer Feedback: Correct because the build-up approach involves 'building up' the required return on equity, beginning with the risk-free rate, and then adding relevant risk premia to account for various risk considerations:

$$r_e = r_f + \text{ERP} + \text{SP} + \text{SCRCP}$$

where SP is a size premium and SCRCP is a specific-company risk premium.

$$\text{Thus, } r_e = 3.5\% + 6.0\% + 4.5\% + 5.0\% = 19.0\%.$$

☐ 22.0%.

Incorrect Answer Feedback: Incorrect because this answer choice uses the expanded CAPM model instead of the build-up approach:

$$r_e = r_f + \beta_{\text{peer}} \text{ERP} + \text{SP} + \text{SCRCP} = 3.5\% + 1.5 \times 6.0\% + 4.5\% + 5.0\% = 22.0\%$$

13 Multiple Choice 0 / 1 point

An analyst is estimating the equity risk premium (ERP) for Trenton Company using the country spread model. Trenton is headquartered in Country 1, a developed market, which has an ERP of 4%. Trenton derives 30% of its sales in Country 2, a developing market with a sovereign yield spread of 3.1%. The analyst collects the following information:

	Country 1	Country 2
Standard deviation of equity market returns	2.5%	4.2%
Standard deviation of bond market returns	2.3%	2.8%

Using the Damodaran model, the analyst's estimated ERP for Trenton should be *closest* to:

☐ 5.01%.

Incorrect Answer Feedback: Incorrect because the calculation has used the standard deviations of the developed market (rather than the developing (local) market).

ERP = ERP for the developed market + ($\lambda \times$ country risk premium)

λ is the is the level of exposure of the company to the local country = 30%, percentage of sales in the developing market.

Country risk premium = Sovereign yield spread $\times (\sigma_{\text{Equity}}/\sigma_{\text{Bond}})$

σ_{Equity} is the volatility of the developed country's equity market = 2.5% and σ_{Bond} is the volatility of the local developed country's bond market = 2.3%.

ERP = 4% + $[0.3 \times 3.1\% \times (2.5\%/2.3\%)] = 5.011\% \approx 5.01\%$

☒ 5.40%.

Correct Answer Feedback: Correct because under the Damodaran model, the ERP is approximately 5.40%, as follows:

ERP = ERP for the developed market + ($\lambda \times$ country risk premium)

λ is the is the level of exposure of the company to the local country. In this case, the level of exposure is 30%, which is percentage of sales derived in the developing market.

Country risk premium = Sovereign yield spread $\times (\sigma_{\text{Equity}}/\sigma_{\text{Bond}})$

σ_{Equity} is the volatility of the local (developing) country's equity market = 4.2% and σ_{Bond} is the volatility of the local (developing) country's bond market = 2.8%.

ERP = 4% + $[0.3 \times 3.1\% \times (4.2\%/2.8\%)] = 5.395\% \approx 5.40\%$

☐ 8.65%.

Incorrect Answer Feedback: Incorrect because the calculation omits the required exposure (λ) of 30% based the percentage level of sales in the developing country.

ERP = ERP for the developed market + country risk premium

Country risk premium = Sovereign yield spread $\times (\sigma_{\text{Equity}}/\sigma_{\text{Bond}})$

σ_{Equity} is the volatility of the local (developing) country's equity market = 4.2% and σ_{Bond} is the volatility of the local (developing) country's bond market = 2.8%.

ERP = 4% + $[3.1\% \times (4.2\%/2.8\%)] = 8.65\%$

An analyst researches Trenson, a private company whose total debt consists of a straight bond issue that pays interest semiannually and has a residual maturity of seven years. To estimate Trenson's cost of debt, the analyst gathers the following information on two straight bonds issued by companies that have similar credit quality to Trenson and pay interest semiannually:

	Coupon Rate	Remaining Maturity	Current Price (Per 100 of Par Value)
Bond 1	4.8%	5 years	102.34
Bond 2	6.4%	8 years	95.67

Based on matrix pricing, the estimated yield to maturity of the Trenson bond is *closest* to:

☒ 5.70%.

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Incorrect Answer Feedback: Incorrect because 5.70% is the average YTM of the 5-year and 8-year bonds, rather than the interpolated YTM of the 7-year bond. Using these YTM from the correct answer, the average YTM is $(4.275\% + 7.119\%)/2 = 5.697\% \approx 5.70\%$

☐ 5.87%.

Incorrect Answer Feedback: Incorrect because it uses the coupon rates instead of the YTM of the 5-year and 8-year bonds in calculating the YTM of the 7-year bond. The coupon rates for the 5- and 8-year bonds are 4.8% and 6.4%, respectively.

The estimated annual incremental in the coupon rate is $(6.4\% - 4.8\%)/(8 - 5) \approx 0.533\%$.

Using linear interpolation, the estimation of the YTM of the 7-year Trenson bond is: $4.8\% + ((7 - 5) \times 0.533\%) = 5.867\% \approx 5.87\%$

☐ 6.17%.

✓

Correct Answer Feedback: Correct because this is the YTM of a bond with 7 years left to maturity using matrix pricing method. Using a financial calculator (with semi-annual payments):

YTM for Bond 1:

$N = 5 \times 2 = 10$; $PV = -102.34$; $PMT = 4.8/2 = 2.4$; $FV = 100$; $CPT\ I/Y = 2.1376\%$. $2.1376\% \times 2 \approx 4.275\%$.

YTM for Bond 2:

$N = 8 \times 2 = 16$; $PV = -95.67$; $PMT = 6.4/2 = 3.2$; $FV = 100$; $CPT\ I/Y = 3.5596\%$. $3.5596\% \times 2 \approx 7.119\%$.

The estimated annual incremental in the yield is $(7.119\% - 4.275\%)/(8 - 5) = 0.948\%$.

Using linear interpolation, the estimation of the YTM of the 7-year maturity Trenson bond is: $4.275\% + ((7 - 5) \times 0.948\%) = 6.171\% \approx 6.17\%$

15 Multiple Choice 1 / 1 point

Which of the following is **not** included in the build-up approach for estimating the required return on equity for private companies?

☒ Beta



Correct Answer Feedback: Correct because an industry beta is a factor in the extended CAPM model, not the build-up model. The extended CAPM model for estimating the required return on equity (cost of equity) is: $r_e = r_f + \beta_{\text{peer}} (\text{ERP}) + \text{SP} + \text{IP} + \text{SCRCP}$, where

r_f = the risk-free rate;

β_{peer} = the industry beta;

ERP = equity risk premium;

SP = size premium;

IP = industry premium; and

SCRCP = specific company risk premium.

☐ A size premium

Incorrect Answer Feedback: Incorrect because a size premium is included in the build-up model. The build-up model for estimating the required return on equity (cost of equity) is calculated as:

$r_e = r_f + \text{ERP} + \text{SP} + \text{SCRCP}$, where

r_f = the risk-free rate;

ERP = equity risk premium;

SP = size premium; and

SCRCP = specific company risk premium.

☐ The risk-free rate

Incorrect Answer Feedback: Incorrect because the risk-free rate is included in the build-up model. The build-up model for estimating the required return on equity (cost of equity) is calculated as:

$r_e = r_f + \text{ERP} + \text{SP} + \text{SCRCP}$, where

r_f = the risk-free rate;

ERP = equity risk premium;

SP = size premium; and

SCRCP = specific company risk premium.